

PAWS 2026: QUATERNION ALGEBRAS

PROBLEM SET 3

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We hope you are enjoying the problems so far and working through the exercises in the lecture notes as well! Here are some new problems based on Lecture 3, giving you an opportunity to explore some of the ideas about central simple algebras and associated conics.

We would love to hear how you are finding these problems! Let us know if you would like us to increase or decrease the difficulty, or if there are particular types of problems you would like to see more of. You can share your thoughts with us on Discord or during our virtual sessions.

- (1) Let $a, b \in F^\times$ be such that $a + b = c^2$ for some $c \in F$. Show that $(a, b)_F$ is split.
- (2) Consider the conic $C : 2x^2 - 7y^2 = 1$. Notice that $P = (2, 1)$ is a $\mathbb{Q}$ -rational point on C .
 - (a) Consider the family of lines

$$y - 1 = t(x - 2), \quad t \in \mathbb{Q},$$

passing through P . Find the second point of intersection of each line with C , and express its coordinates as rational functions of t .

- (b) How many $\mathbb{Q}$ -rational points do you think C has? Why?
- (3) Let p be a prime. Determine all primes p for which the quaternion algebra $(-1, p)_{\mathbb{Q}}$ is split.

Hint: You may use the fact that an odd prime p can be written as a sum of two squares of integers if and only if $p \equiv 1 \pmod{4}$.
- (4) In the proof of Theorem 3.6, we used the following result:

Theorem 1. *The multiplicative group $\mathbb{F}_q^\times$ of a finite field $\mathbb{F}_q$ is a cyclic group of order $q - 1$.*

This exercise outlines a proof of this result.

- (a) Show that $\mathbb{F}_q^\times$ is an abelian group of order $q - 1$ under multiplication.
- (b) The case $q = 2$ is quite simple, so let us assume $q > 2$. The *structure theorem for finite abelian groups*¹ gives a direct product decomposition

$$\mathbb{F}_q^\times \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}, \quad n_r \mid n_{r-1} \mid \cdots \mid n_1,$$

where $n_i > 1$ for all $i = 1, \dots, r$ and $r \in \mathbb{Z}_+$.

- (c) Under this isomorphism, regard each direct factor $\mathbb{Z}/n_i\mathbb{Z}$ as a subgroup of $\mathbb{F}_q^\times$. Verify that all elements of this subgroup satisfy the polynomial $X^{n_i} - 1$.
- (d) Using the divisibility relations among the n_i , show that every element of $\mathbb{F}_q^\times$ is a root of $X^{n_1} - 1$. Deduce that $q - 1 \leq n_1$. *Hint:* How many roots can a nonzero polynomial of degree n_1 have over a field?
- (e) On the other hand, the direct product decomposition gives

$$q - 1 = n_1 n_2 \cdots n_r.$$

Combine this with part (c) to show that $r = 1$, and conclude that $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

¹Review this if you forgot what this says!

- (5) In this exercise, we verify another important idea used in the proof of Theorem 3.6. Let p be an odd prime and let $a, b \in (\mathbb{Z}/p\mathbb{Z})^\times$.

(a) Show that the set

$$S_1 = \{x^2 \pmod{p} \mid x \in \mathbb{Z}/p\mathbb{Z}\}$$

has exactly $\frac{p+1}{2}$ elements.

(b) Show that the set

$$S_2 = \{b + ay^2 \pmod{p} \mid y \in \mathbb{Z}/p\mathbb{Z}\}$$

also has exactly $\frac{p+1}{2}$ elements.

(c) Use the pigeonhole principle to conclude that $S_1 \cap S_2 \neq \emptyset$, and deduce that the congruence

$$x^2 - ay^2 \equiv b \pmod{p}$$

always has a solution for any choice of units a and b .

- (6) Let p and q be distinct odd primes, and consider the conic $C : px^2 + qy^2 = z^2$ associated to the quaternion algebra $(p, q)_{\mathbb{Q}}$. We say that a point of C (or one of the analogously defined conics below) is a **good point**² if it is nontrivial (i.e. not all coordinates are 0) and z is invertible.

(a) For an odd prime $\ell \neq p, q$, show that $C_\ell : \bar{p}x^2 + \bar{q}y^2 = z^2$, the reduction of C modulo ℓ , always has a good point over $\mathbb{F}_\ell$ (Hint: follow the proof of Theorem 3.6 from the notes!).

(b) For $\ell = p$, show that C_ℓ has a good point over $\mathbb{F}_p$ if and only if q is a square modulo p (and analogously for $\ell = q$).

(c) For $\ell = 2$, we make a slight modification and define $C_2 : \bar{p}x^2 + \bar{q}y^2 = z^2$ modulo 4 (rather than 2). Show that C_2 has a good point over $\mathbb{Z}/4\mathbb{Z}$ if and only if at least one of p or q is equivalent to 1 modulo 4.

(d) Recall from quadratic reciprocity that for odd primes p, q , we have that

- If at least one of p or q is equivalent to 1 modulo 4, then p is a square modulo q if and only if q is a square modulo p .
- If both p or q are equivalent to 3 modulo 4, then either p is a square modulo q or q is a square modulo p , but not both.

Use this, together with the above results, to show that the set of primes

$$S = \{\ell \text{ prime} \mid C_\ell \text{ does NOT have good points}\}$$

is a finite set with even cardinality.

- (7) In this exercise we will prove a formula for the number of points on affine conics over finite fields. This gives a refinement of exercise 5.

Let p be an odd prime, n a positive integer, and $q = p^n$. Define the quadratic character χ to be the map

$$\chi : \mathbb{F}_q \rightarrow \{-1, 0, 1\}$$

$$x \mapsto \begin{cases} 1 & x \neq 0 \text{ and } x = y^2 \text{ for some } y \in \mathbb{F}_q \\ -1 & x \neq 0 \text{ and } x \text{ is not a square in } \mathbb{F}_q \\ 0 & x = 0. \end{cases}$$

An equivalent definition of χ is that for a fixed $x \in \mathbb{F}_q$, the equation $x = y^2$ has $1 + \chi(x)$ solutions in $\mathbb{F}_q$. You may use that χ is a multiplicative function, so $\chi(xy) = \chi(x)\chi(y)$.

(a) For a fixed $d \in \mathbb{F}_q^\times$, show that the equation

$$x^2 - y^2 = d$$

has exactly $q - 1$ solutions $(x, y) \in \mathbb{F}_q^2$. Hint: factorize $x^2 - y^2$.

²This definition will become more natural when phrased in the language of the p -adic numbers, which will be defined in Lecture 4.

(b) Deduce that

$$\sum_{x \in \mathbb{F}_q} \chi(x^2 - d) = -1.$$

Hint: use the equivalent definition of χ .

(c) Let $a, b \in \mathbb{F}_q^\times$, and let $C(a, b)$ be the conic defined by the equation

$$ax^2 + by^2 = 1.$$

Let $N(a, b)$ be the number of solutions $(x, y) \in \mathbb{F}_q^2$ to this equation. Show that $N(a, b) = q - \chi(-ab)$. Hint: use a similar sum as in part b.

(8) In this exercise we compute the number of elements of $M_2(\mathbb{F}_q)$ with a given determinant. Since every quaternion algebra over $\mathbb{F}_q$ is split, this is equivalent to computing the number of elements of a quaternion algebra over $\mathbb{F}_q$ with given norm.

(a) Show that

$$|\mathrm{GL}_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q).$$

(b) Show that the determinant map

$$\det : \mathrm{GL}_2(\mathbb{F}_q) \rightarrow \mathbb{F}_q^\times$$

is surjective, and that for all elements $x, y \in \mathbb{F}_q^\times$, their preimages satisfy $|\det^{-1}(x)| = |\det^{-1}(y)|$.

(c) Conclude that for all $x \in \mathbb{F}_q^\times$, we have

$$|\det^{-1}(x)| = (q + 1)(q^2 - q).$$

(d) Show that the number of matrices in $M_2(\mathbb{F}_q)$ with zero determinant is $q^3 + q^2 - q$. How many nonzero zero divisors does $M_2(\mathbb{F}_q)$ have?